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Exploring the usage behavior of generative artificial intelligence: a case study of ChatGPT with insights into the moderating effects of habit and personal innovativeness

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Abstract
This study, using ChatGPT as a case, based on the Unified Theory of Acceptance and Use of Technology 3 (UTAUT3) model, explores university students' attitudes, Behavioral Intentions (BI), and influencing factors towards Generative Artificial Intelligence (GenAI). It focuses on two crucial moderating variables: Habit (H) and Personal Innovativeness (PI). The goal is to uncover the mechanisms through which H and PI influence GenAI BI, providing a deeper understanding of GenAI Usage Behavior (UB). This study conducted a comprehensive survey of 510 college students, utilizing SPSS 26.0 and AMOS 24.0 for data analysis. Results reveal significant positive impacts of Performance Expectations (PE), Effort Expectations (EE), Social Influence (SI), and Hedonic Motivation (HM) on students' BI to use ChatGPT, except for Price Value (PV). Additionally, BI significantly influences actual UB. Moderation analysis shows H negatively moderates the BI-UB relationship, indicating its inhibitory effect, while PI positively moderates this relationship, emphasizing the innovative role in promoting GenAI usage. These findings contribute to a theoretical understanding of GenAI UB and offer practical product design and promotion insights.

Keywords GenAI · ChatGPT · Usage behavior · UTAUT3 · Habit · Personal innovativeness

Introduction

GenAI refers to a set of artificial intelligence techniques based on technologies such as Generative Adversarial Networks and large-scale pre-trained models. Through learning and recognition from existing data, GenAI possesses the capability to generate relevant content with appropriate generalization. The significance of GenAI for human society and artificial intelligence is monumental. In the short term, GenAI has altered fundamental productivity tools, while in the medium term, it is poised to transform societal production relations. In the long term, GenAI is anticipated to drive a qualitative breakthrough in overall societal productivity. Within this transformation of productivity tools, production relations, and productivity, the factor of production—data value is greatly amplified. GenAI elevates data elements to the position of a core resource in the era, thereby contributing to a certain extent to an accelerated digital transformation of society.

In the context of the digital age, GenAI as a significant technological approach has begun to be widely applied in various fields such as education, law, media, and library services, sparking discussions on a range of social, ethical, and legal issues. Li et al. (2024) found through a survey of 1,190 college students that most students understand GenAI and often use it for scientific research activities, course learning, showing its widespread application in education. Han (2023) focused on the challenges of GenAI in copyright issues, emphasizing the importance of resolving copyright ownership and protecting creative content. Zhang (2023)

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investigated how news organizations are actively integrating GenAI into their workflows, reflecting the positive application of GenAI in the media industry.

In the field of library reading services, Wu and Mao (2023) analyzed GenAI to show that it has driven the transformation of library reading services toward digitalization, networking, and even intelligence. Xu (2023), from the perspective of industrial applications, comprehensively sorted out GenAI's development and application status, revealing the opportunities and challenges GenAI brings in multiple industrial fields.

Shang (2023), Song et al. (2023), and Yuan (2023) discussed the management issues of GenAI from the perspectives of risk governance, governance strategies, and responsibility capability, respectively. They pointed out that it is necessary to build a practical regulatory framework to face the challenges of GenAI in national security, social ethics, and private rights. Yuan and Zheng (2023), through analyzing the application of artificial intelligence in architectural design, proposed new architectural design models, illustrating the innovative applications of GenAI in professional fields.

Through a detailed analysis of existing literature, it can be seen that although the application of GenAI technology has provided new opportunities for various industries, promoting the improvement of productivity and service quality, it has also triggered a series of issues, such as copyright disputes, ethical considerations, privacy protection, and responsibility delineation.

According to data from NerdyNav, ChatGPT is one of GenAI's concrete applications. It has 180.5 million users, of which 100 million are active weekly. In January 2024, ChatGPT received 2.3 billion website visits, with 2 million developers currently using its Application Program Interface(API). This has elevated the content generation capability of intelligent question-and-answer bots to new heights.

The content produced by ChatGPT finds widespread applications in education, research, consultation, entertainment, and various other activities. Michael (2024) underscores the dual-edged nature of ChatGPT in educational settings, positioning it as both an opportunity and a threat. The study hinged on a constructivist-interpretative perspective and examined academics' apprehensions and strategic responses at a faculty of education, shedding light on the nuanced views towards GenAI-based conversational tools.

However, with the in-depth utilization of ChatGPT across diverse domains, research has surfaced, highlighting several drawbacks in its content generation. Issues such as conversation length limitations, ethical and moral concerns, intellectual property considerations, data biases, low credibility, and data pollution have been identified. These challenges contribute to a decrease in user trust, thereby affecting the further adoption of GenAI (Mao et al., 2023). This poses challenges to realizing GenAI's value and future practical applications.

Current research on GenAI primarily focuses on discussions on developing text and image generation tasks and related models (Cao et al., 2023). ChatGPT, developed by the OpenAI team, stands out as a representative GenAI model. Research on the willingness to use ChatGPT mainly centers on its benefits, limitations, ethical considerations (Ali, 2023), user perceptions (Miyazaki et al., 2023), and the impact on user willingness using principal component analysis (Li, 2023). There is limited specific research on the Behavioral Intentions (BI) to use ChatGPT and whether there are moderating variables influencing Usage Behavior (UB).

Therefore, this article takes ChatGPT as an example and constructs a model based on the UTAUT3 model to explore the factors influencing university students' intention to use GenAI. The aim is to investigate the causal relationships and associated paths among various factors. The goal is to provide a reference basis for clarifying design directions and enhancing usage value for other GenAI products.

Theoretical foundation and hypotheses

Theoretical foundation

Since the late 1980s, theoretical models for assessing the acceptance of information systems have been under development. Each theory introduces different dimensions to explore individuals' acceptance and use of technology. According to Venkatesh et al. (2003), the UTAUT was proposed based on an experimental combination of eight different models adapted from psychological and sociological theories. It aims to explain the factors influencing organizational employees' acceptance and use of Information and Communication Technology (ICT). It has primarily been employed in previous research to explain the acceptance and use of new technologies.

The eight models and theories include Theory of Reasoned Action (TRA), Technology Acceptance Model (TAM), Motivational Model (MM), Theory of Planned Behavior (TPB), Combined TAM and TPB (C-TAM-TPB), PC Utilization Model (MPCU), Innovation Diffusion Theory (IDT), and Social Cognitive Theory (SCT). The conceptual model of UTAUT is built upon four latent variables: performance expectancy, effort expectancy, facilitating conditions, and social influence, serving as determinants of usage behavior. Venkatesh et al. (2012) proposed the UTAUT2 with the intention of exploring factors influencing technology acceptance and usage from a consumer perspective. It integrates

three new variables: Hedonic Motivation (HM), Price Value (PV), and Habit (H).

Comparatively, Farooq et al. (2017) introduced the UTAUT3 model on this basis, incorporating the concept of Personal Innovativeness (PI). In the IT context, PI refers to individuals' perceived tendency or personal disposition reflecting their inclination to experiment and independently adopt new advancements in information technology. In other words, PI can be conceptualized as the willingness or adventurous inclination to adopt the latest technological products, potentially related to trying out new features and advancements in the IT domain (Agarwal, 1998). The UTAUT3 model was initially tested in a higher education environment (Gunasinghe et al., 2019a).

Performance expectancy(PE) and behavioral intention(BI)

Venkatesh et al. (2003) pointed out that PE refers to the degree to which an individual believes that adopting a specific technology will provide advantages in carrying out certain activities. Venkatesh et al. (2012) modified the previous definition to the extent to which technology adoption and use benefit customers when performing certain activities, and users believe that targeted technology will enhance their performance to gain work-related benefits. San Martín and Herrero (2012) used the UTAUT model to explain the positive correlation between PE and travelers' intention to visit. Im et al. (2011) claimed that PE is an essential and potential predictor of BI to adopt online travel.

Factors related to performance are potential predictors of individual intentions to use or adopt new technologies (Schaupp et al., 2010). Similarly, from an educational perspective, PE refers to academicians' belief that e-learning will assist them in completing job tasks (Gunasinghe et al., 2019a).

In this study, PE refers to university students' belief that using ChatGPT can effectively and efficiently help them accomplish their learning tasks. Therefore, the following hypothesis is proposed:

Hypothesis 1 PE has a significant positive impact on university students' BI when using ChatGPT.

Effort expectancy(EE) and behavioral intention(BI)

Venkatesh et al. (2003, 2012) posit that EE is the ease with which an individual expects to use a specific system, believing that the interaction with the technology is either troublesome or trouble-free and perceiving that interaction with the target technology is smooth. Godoe and Johansen (2012) assert that customers always choose user-friendly technologies that offer maximum efficiency. If a new technology is easy to understand and requires less effort to acquire knowledge about it, customers are more likely to adopt it. From an educational perspective, EE is the belief among academicians in the ease or convenience of using an e-learning platform (Gunasinghe et al., 2019b). If the use or adoption of technology is straightforward, it contributes to the convenience of students and teachers.

In this study, EE refers to the belief among university students that they can quickly learn and proficiently master the use of ChatGPT. Therefore, the following hypothesis is proposed:

Hypothesis 2 EE has a significant positive impact on university students' BI when using ChatGPT.

Social Influence(SI) and behavioral intention(BI)

Ajzen (1991) posited that SI is the social pressure that compels individuals to take or not take specific actions. Venkatesh et al. (2012) consider SI as the degree to which an individual believes that the society in which they exist requires them to use a particular technology and its importance. Tan (2013) suggests that the primary aim of SI is to highlight the perspectives of crucial figures in our lives, including family members, friends, and colleagues. Scholars like Leong et al. (2013) have found a positive correlation between consumers' SI and BI. Research indicates that social pressure from peers or higher authorities significantly influences consumers' adoption of new technology (Gonzalez et al., 2012).

In this study, SI refers to the impact on university students from authoritative scholars, friends, and family before using ChatGPT. Therefore, the following hypothesis is proposed:

Hypothesis 3 SI has a significant positive impact on university students' BI when using ChatGPT.

Hedonic motivation(HM) and behavioral intention(BI)

Venkatesh et al. (2012), in the UTAUT2 model, highlighted HM as the pleasure or enjoyment derived from using a specific technology, and it has been proven to play a significant role in determining the acceptance and use of technology. In the field of information technology, research has found that HM has a direct impact on BI. Scholars like N. Shaw (2019) discovered that HM significantly influences users' willingness to use mobile commerce. HM has also been identified as a crucial determinant in adopting information technology in the consumer domain. Li (2016) found that HM positively impacts university students' willingness to use fast-moving consumer goods brand apps.

In this study, HM is reflected in the pleasure obtained by university students when using ChatGPT. The features of ChatGPT, such as rapid semantic content generation and code generation, can provide enjoyment to university students. This anticipated pleasure is expected to increase the willingness of university students to use ChatGPT. Therefore, the following hypothesis is proposed:

Hypothesis 4 HM has a significant positive impact on university students' BI when using ChatGPT.

Price value(PV) and behavioral intention(BI)

Dodds et al. (1991) defined PV as the perceived balance between the benefits of using a specific technology and the amount spent on using that technology. It involves comparing the costs one may incur in choosing to use the technology with the various economic benefits obtained. Scholar Yang and Zhang (2020), in a study on the impact of e-book users' BI, found that PV has a positive impact on BI. PV also demonstrates exemplary performance in the user's BI of car-sharing services (Shi, 2017).

In this study, when university students perceive that the experiential and cost-effectiveness benefits of using ChatGPT outweigh the costs they need to incur, their willingness to use ChatGPT is expected to increase. Therefore, the following hypothesis is proposed:

Hypothesis 5 PV has a significant positive impact on university students' BI when using ChatGPT.

Moderating effect of personal innovativeness(PI)

In Information Technology (IT), PI undeniably correlates with the eventual adoption of technologies. It refers to an individual's willingness to experiment with new information technology, representing a relatively stable individual trait. Additionally, Thatcher and Perrewé (2002) emphasizes that PI is a stable, context-specific characteristic with a strong tendency to influence the adoption and acceptance of IT.

The "IT domain's PI" differs from the concept of "Innovativeness" proposed by Rogers (1995) in IDT. The innovativeness structure used in IDT measures the standard of adopting technology early compared to others. In contrast, the "IT domain's PI" is an individual attribute instilling a desire and openness to try new advancements in the IT domain. It signifies the tendency of academic users to independently make innovative decisions without the need for communication with others (Midgley, 1978). Ratten (2014) suggests that users in China and the United States with higher PI are more inclined to adopt cloud computing technology. As a moderating variable, PI also promotes users' willingness to use it continuously. Mao (2017) found that PI positively moderates the relationship between customer perceived value and continuous usage intention in intelligent wearable technology devices.

In this study, when university students intend to use ChatGPT, strong PI characteristics are expected to contribute to a more sustained usage pattern. Therefore, the following hypothesis is proposed:

Hypothesis 6 PI positively moderates the relationship between university students' BI to use ChatGPT and their actual UB.

Moderating effect of habit(H)

Venkatesh et al. (2012) have pointed out that individuals, due to prior experiences, may unconsciously or automatically engage in behaviors, with H reflecting these experiences. However, H is not solely formed through experiences; even if an H is established, it may require minimal favorable conscious motivation to sustain the behavior over the long term (Gardner et al., 2020). H is a cognitive process where situations automatically trigger impulses to take action due to the activation of context-action associations learned through repeated performances (Lally et al., 2010). In stable and familiar environments, H tends to dominate the regulation of BI (Neal et al., 2012).

In studies related to the use of information technology, Bhattacherjee (2015) and Limayem (2011) found that H, as a moderating variable, negatively influenced the relationship between BI and UB. Similarly, in Moody and Siponen's (2013) research on individuals' internet UB at work, H, as a moderating variable, had a negative impact.

In this study, university students may have formed H related to the use of other GenAI outside the ChatGPT environment. The similarity in behavior between ChatGPT and other GenAI might weaken the sustained UB under the moderating effect of H when university students express the intention to use ChatGPT. Therefore, the following hypothesis is proposed:

Hypothesis 7 H negatively moderates the relationship between university students' BI to use ChatGPT and their actual UB.

Behavioral intention(BI) and usage behavior(UB)

According to Venkatesh et al. (2003, 2012), BI is the user's inclination to use self-service technology (SST). BI has been repeatedly validated in the technology adoption process and has become the most influential determinant of UB. Furthermore, previous literature (Jaruwachirathanakul

& Fink, 2005; Martins et al., 2014) has recognized BI as a decisive predictive factor for actual UB. Therefore, the following hypothesis is proposed:

Hypothesis 8 BI has a significant positive impact on university students' use of ChatGPT.

Based on the theoretical overview and research hypotheses presented above, the following hypothetical model is proposed in Fig. 1

Research methodology

In the theoretical model proposed in this study, each variable used is not directly observable or measurable. Therefore, Structural Equation Modeling (SEM) was chosen as the analytical tool. To ensure the reliability and validity of the scales used, an initial scale was formed based on the content of the study, drawing inspiration from scales used by various scholars both domestically and internationally. Considering cultural differences, semantic modifications were made to the scales to align with Chinese language conventions and cultural backgrounds. Further refinement was done through interviews with several respondents. A pilot survey was conducted to optimize the effectiveness and feasibility of the questionnaire. After the steps above and analysis of pilot survey data, the final test questionnaire was formulated. The questionnaire consists of two parts: the first part investigates the demographic characteristics of the subjects, and the second part addresses questions related to the hypothesized variables.

This study targeted Chinese university students as the survey participants. The questionnaire was distributed through research institutions, survey platforms, emails, etc. A total of 578 questionnaires were distributed, and after excluding ineligible ones, a total of 510 valid questionnaires were collected, resulting in a questionnaire distribution and collection efficiency of 88.2%. Among the participants, 57.1% were male, and 42.9% were female. Regarding the age distribution of respondents, the majority fell into the 18–25 age group and the 26–30 age group, accounting for a total of 75.5%. Regarding educational background, the majority held a bachelor's degree, constituting 62.5%.

The questionnaire utilized a Likert 5-point scale for assessment, where "1" indicated strongly disagree, "3" represented neutral, and "5" indicated strongly agree.

Data analysis and results

Reliability analysis

Reliability, or reliability coefficient, refers to the consistency or stability of the measurement results obtained using a measuring tool. We generally use the α coefficient, also known as Cronbach's Alpha, to assess the reliability of the questionnaire. The higher the α coefficient, the greater the reliability of the questionnaire, indicating higher credibility and stability. Bagozzi and Yi (1988) proposed that the questionnaire reliability is considered very good when the α coefficient falls between 0.80 and 0.90.

Additionally, we utilize Cronbach's Alpha coefficient after item deletion to purify the measurement entries. If

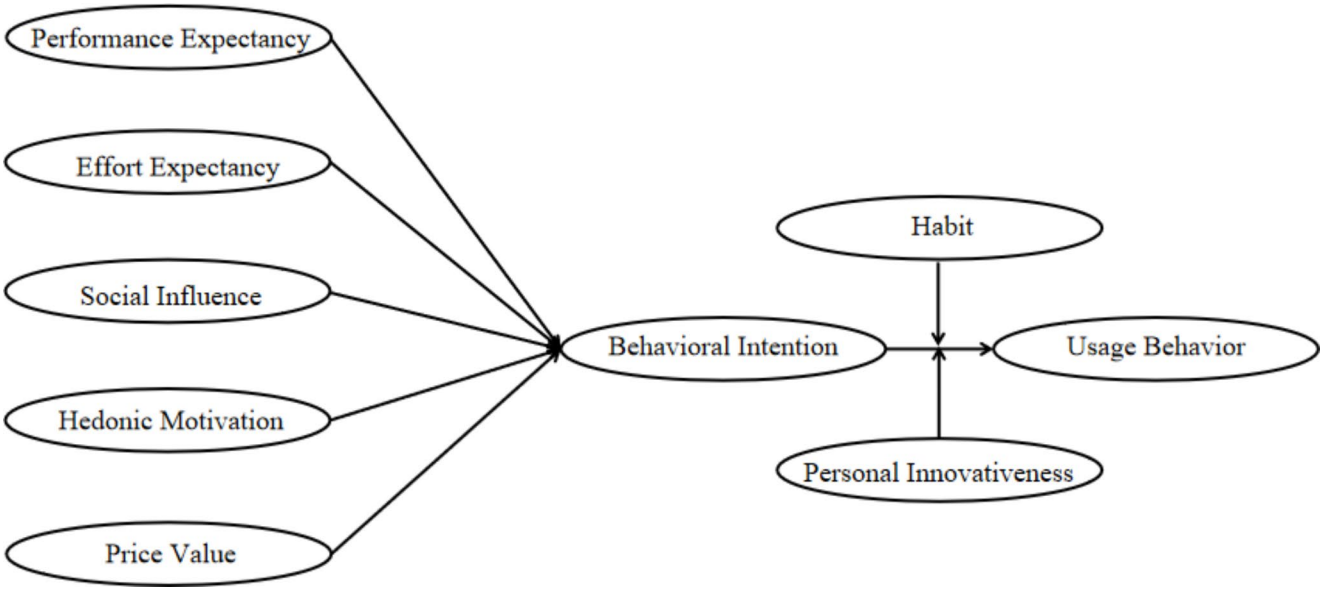

Fig. 1 Hypothetical Model

deleting a specific measurement item increases Cronbach's Alpha coefficient, it indicates that removing that item contributes to improving the internal consistency of the measurement tool. The Corrected Item-Total Correlation (CITC) is employed to assess the statistical relationship between each measurement item (question or problem) and the overall measurement tool (total score). When CITC values range from −1 to 1, higher values indicate a closer relationship between the measurement item and the overall measurement tool.

As shown in Table 1, Cronbach's Alpha coefficients for each variable are all above 0.8, CITC values are all greater than 0.5, and deleting items does not increase Cronbach's Alpha values. Therefore, this indicates that the scale is reliable.

Validity analysis

Exploratory factor analysis

This study employed SPSS 26.0 software to conduct exploratory factor analysis to assess the construct validity of the questionnaire. Initially, the Kaiser-Meyer-Olkin (KMO) measure of sample adequacy and Bartlett's test of sphericity were used to evaluate whether the data were suitable for factor analysis. Following Kaiser's (1974) guideline, a KMO above 0.90 suggests that the scale is highly suitable for factor analysis, while a KMO between 0.8 and 0.9 indicates that it is suitable for factor analysis. Bartlett's test of sphericity aims to examine whether the covariance matrix among observed variables is a unit matrix and if the correlations between variables are zero. Suppose the result of Bartlett's test is significant, meaning the p-value is below the significance level (usually 0.05). In that case, it indicates a correlation among observed variables, making it appropriate for factor analysis.

Bartlett's test of sphericity for the correlation matrix of the questionnaire yielded a significant result with Bartlett's $\chi 2 = 6710.107$, $p < 0.001$. This implies the presence of common factors among the questionnaire items, justifying the need for factor analysis. Simultaneously, the KMO measure was computed, resulting in KMO = 0.855, indicating suitability for factor analysis.

Table 2 shows that this study employed the commonly used principal component analysis method, extracting factors based on the criterion of eigenvalues greater than 1. After extracting factors from the 27 questionnaire items, nine factors were retained, accounting for a cumulative explained variance of 76.119%. This indicates that these factors can explain a significant portion of the total variance.

Table 1 Variable reliability and Validity Analysis

Variable	Item	CITC	Deletion Cronbach's Alpha	Cronbach's Alpha	Factor Loadings	CR	AVE
PE	PE1	0.688	0.717	0.813	0.824	0.814	0.594
	PE2	0.648	0.760		0.746		
	PE3	0.659	0.750		0.739		
EE	EE1	0.696	0.719	0.816	0.791	0.818	0.599
	EE2	0.666	0.753		0.750		
	EE3	0.648	0.769		0.781		
SI	SI1	0.701	0.782	0.840	0.818	0.841	0.639
	SI2	0.678	0.803		0.760		
	SI3	0.737	0.747		0.818		
HM	HM1	0.734	0.756	0.842	0.834	0.844	0.643
	HM2	0.696	0.790		0.783		
	HM3	0.693	0.796		0.788		
PV	PV1	0.673	0.797	0.836	0.753	0.838	0.633
	PV2	0.682	0.788		0.759		
	PV3	0.741	0.729		0.870		
H	H1	0.684	0.765	0.828	0.775	0.828	0.616
	H2	0.684	0.764		0.789		
	H3	0.690	0.759		0.791		
PI	PI1	0.669	0.793	0.833	0.751	0.834	0.627
	PI2	0.707	0.757		0.813		
	PI3	0.708	0.756		0.810		
BI	BI1	0.666	0.825	0.843	0.745	0.847	0.649
	BI2	0.752	0.740		0.863		
	BI3	0.711	0.779		0.805		
UB	UB1	0.741	0.774	0.851	0.838	0.853	0.659
	UB2	0.696	0.817		0.784		
	UB3	0.728	0.786		0.812		

Table 2 Exploratory Factor Analysis Results

Item	1	2	3	4	5	6	7	8	9
SI3	**0.889**	0.059	0.041	0.040	0.093	−0.015	0.074	0.052	−0.015
SI2	**0.825**	0.051	0.009	0.065	0.057	0.139	0.061	0.182	0.063
SI1	**0.810**	0.126	0.008	0.045	0.126	0.173	0.207	−0.051	0.106
HM1	0.065	**0.843**	0.091	0.032	0.163	0.121	0.097	0.110	0.059
HM2	0.112	**0.829**	0.088	0.075	0.147	0.080	0.104	0.027	0.048
HM3	0.056	**0.808**	0.181	0.052	0.171	0.126	0.098	0.042	0.007
PI3	0.017	0.105	**0.850**	0.041	0.055	0.088	0.137	0.027	0.025
PI2	−0.017	0.159	**0.846**	0.017	0.092	0.085	0.086	0.064	0.001
PI1	0.057	0.068	**0.829**	−0.043	0.062	0.077	0.119	0.010	0.106
H1	0.028	0.067	0.001	**0.833**	−0.017	0.094	0.068	0.088	0.169
H3	0.049	−0.005	−0.009	**0.828**	0.027	0.144	0.148	0.068	0.132
H2	0.072	0.093	0.021	**0.815**	0.000	0.148	0.153	0.043	0.130
PV3	0.141	0.141	0.105	0.016	**0.851**	0.115	0.019	0.088	0.083
PV2	0.112	0.106	0.040	−0.042	**0.842**	0.056	0.049	0.032	0.063
PV1	0.014	0.219	0.074	0.071	**0.821**	0.043	0.090	−0.008	0.044
BI2	0.117	0.150	0.078	0.109	0.119	**0.802**	0.151	0.168	0.252
BI3	0.076	0.154	0.107	0.162	0.086	**0.775**	0.175	0.088	0.206
BI1	0.139	0.091	0.141	0.207	0.060	**0.739**	0.209	0.109	0.120
UB3	0.051	0.137	0.124	0.146	0.069	0.168	**0.812**	0.129	0.151
UB1	0.118	0.088	0.133	0.152	0.049	0.232	**0.799**	0.098	0.150
UB2	0.232	0.122	0.181	0.144	0.070	0.119	**0.763**	0.106	0.142
EE1	−0.010	0.117	0.036	0.050	0.044	0.056	0.128	**0.856**	0.089
EE2	0.103	0.019	0.063	0.056	0.050	0.010	0.054	**0.851**	0.137
EE3	0.114	0.038	0.000	0.113	0.012	0.336	0.110	**0.752**	0.129
PE3	0.069	−0.005	0.019	0.171	0.041	0.098	0.111	0.091	**0.830**
PE1	0.029	0.055	0.037	0.153	0.087	0.204	0.208	0.148	**0.777**
PE2	0.046	0.071	0.092	0.141	0.080	0.204	0.089	0.131	**0.772**
Eigenvalues	2.333	2.324	2.314	2.309	2.285	2.271	2.255	2.235	2.227
Variance%	8.642	8.608	8.570	8.551	8.463	8.411	8.351	8.276	8.247
Total%	8.642	17.250	25.820	34.371	42.834	51.245	59.596	67.872	76.119

All loadings of the nine factors are more significant than 0.7, and the allocation of items after factor rotation aligns with the theoretical structure of the questionnaire. Consequently, the questionnaire exhibits good construct validity.

Confirmatory factor analysis

This study employed AMOS 24.0 software to conduct confirmatory factor analysis on the sample. Confirmatory factor analysis is a statistical analysis of survey data, aiming to examine whether the factor structure model of the scale fits the collected data and whether the indicator variables can effectively serve as measurement variables for latent variables. Maximum likelihood estimation was utilized to estimate the model in this research, and the model fit was assessed through several indices: (1) Chi-square (χ^2) test. The chi-square test is a fundamental measure of model fit, often assessed using the χ^2/df ratio, where smaller values indicate better fit. Generally, when $\chi^2/df < 3$, it suggests a good fit. (2) Root Mean Square Error of Approximation (RMSEA). RMSEA is particularly sensitive to misspecifications in the model and is considered an ideal fit index. A lower RMSEA value, closer to 0, indicates a better model fit. Typically, RMSEA < 0.08 suggests a good fit. (3) Standardized Root Mean Square Residual (SRMR). Ranging from 0 to 1, an SRMR < 0.08 suggests a well-fitted model. (4) Normed Fit Index (NFI), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), and Incremental Fit Index (IFI). Generally, NFI > 0.9, CFI > 0.9, TLI > 0.9, and IFI > 0.9 indicate a good model fit.

Through the relevant data processing, this study's model fit indices, including χ^2/df (1.771), RMSEA (0.039), SRMR (0.037), NFI (0.925), CFI (0.966), TLI (0.958), and IFI (0.966), have all reached the desired values. Therefore, this indicates that the fit of the confirmatory factor analysis model is good.

As shown in Table 1, following Hair et al.'s (2011) recommendations for validity assessment, the factor loadings for each variable are all above 0.5, the Composite Reliability (CR) values are all above 0.7, and the Average Variance Extracted (AVE) values are all above 0.5. Therefore, the questionnaire demonstrates good convergent validity.

Examining Table 3, which presents the correlation coefficients between latent variables and the square root of AVE, this study follows the method proposed by Fornell and Larcker (1981). Specifically, it assesses whether the square root of AVE for each variable is higher than the correlation coefficients between the two variables. The results indicate that the square root of AVE for each variable is greater than the corresponding correlation coefficient, suggesting that the questionnaire exhibits good discriminant validity.

Correlation analysis and discriminant validity

The Pearson correlation coefficient measures the linear relationship between two interval variables. The coefficient ranges from −1 to 1, where a more considerable absolute value is not equal to 0, and a significance level of $p<0.05$ indicates a stronger correlation between the two variables. This study uses the Pearson correlation coefficient to examine the relationships between variables.

From the data in the lower left corner of Table 3, standard deviations and correlation coefficients among the research variables can be observed. BI is significantly positively correlated with PE ($r=0.491$, $p<0.01$), EE ($r=0.371$, $p<0.01$), SI ($r=0.307$, $p<0.01$), HM ($r=0.352$, $p<0.01$), PV ($r=0.255$, $p<0.01$), and UB ($r=0.510$, $p<0.01$). Therefore, the hypotheses receive preliminary support.

This paper additionally utilizes a second method to assess discriminant validity. According to the approach proposed by Henseler et al. (2015), the closer the HTMT values are to 0.85 between pairs of variables, the poorer the discriminant validity. From the data in the upper right corner of Table 3, it can be observed that the HTMT values between BI, EE, H, HM, PE, PI, PV, SI, and UB are all below the testing criterion of 0.85, thereby indicating that the questionnaire exhibits good discriminant validity.

Structural equation modeling analysis

This study employed structural equation modeling for hypothesis testing (Fig. 2). Structural equation modeling is a statistical method based on the covariance matrix of variables, integrating techniques such as factor analysis, multivariate regression analysis, and path analysis to analyze relationships between variables. Through the relevant data processing in this study, the model fit indices, including χ^2/df (2.381), RMSEA (0.052), SRMR (0.052), NFI (0.922), CFI (0.953), TLI (0.943), and IFI (0.953), the fit indices of the model meet the ideal criteria, indicating a good fit of the structural equation model. All other hypotheses are supported through the relevant data processing, except for the non-significant influence of PV on BI ($p=0.600$).

Building upon the structural equation model, we further utilized the bias-corrected non-parametric percentile bootstrap method to estimate the significance of specific indirect effects, with 5000 bootstrap iterations, except for the path PV→BI→UB, where the confidence interval of the indirect effect includes 0 (−0.043, 0.075), indicating no indirect effect of BI between PV and UB, other paths demonstrate the presence of indirect effects through BI.

Moderation effect analysis

This study employed hierarchical regression analysis to examine the moderating effects of H and PI. Initially, all variables were standardized. Subsequently, demographic variables were regressed on the dependent variable. Then, the independent variable was regressed on the dependent variable. Following this, the independent and moderating variables were regressed on the dependent variable. Finally, the interaction term between the independent and moderating variables was introduced in the regression equation. The observation focused on changes in R^2 and p-values. If R^2 significantly increased or the p-value of the interaction term reached the significance level, it would indicate that the variables have moderating effects.

From Table 4, it can be observed that in comparison to Model Three, Model Four exhibits a significant increase in R^2 ($\triangle R^2=0.065$, $p<0.001$) after introducing the interaction terms between BI and H, as well as BI and PI. The interaction term between BI and H significantly negatively

Table 3 Correlation analysis and discriminant validity

	PE	EE	SI	HM	PV	H	PI	BI	UB
PE	**0.771**	0.422	0.228	0.213	0.240	0.483	0.197	0.595	0.500
EE	0.343**	**0.774**	0.243	0.232	0.163	0.275	0.154	0.443	0.384
SI	0.186**	0.201**	**0.799**	0.276	0.291	0.200	0.118	0.367	0.399
HM	0.173**	0.194**	0.231**	**0.802**	0.466	0.203	0.362	0.419	0.388
PV	0.197**	0.136**	0.244**	0.391**	**0.796**	0.074	0.247	0.304	0.251
H	0.397**	0.228**	0.165**	0.170**	0.061	**0.785**	0.079	0.477	0.445
PI	0.160**	0.127**	0.097*	0.306**	0.207**	0.065	**0.792**	0.332	0.402
BI	0.491**	0.371**	0.307**	0.352**	0.255**	0.399**	0.279**	**0.806**	0.602
UB	0.416**	0.322**	0.336**	0.329**	0.212**	0.373**	0.338**	0.510**	**0.812**

*$p<0.05$, **$p<0.01$, Diagonal is the square root of AVE

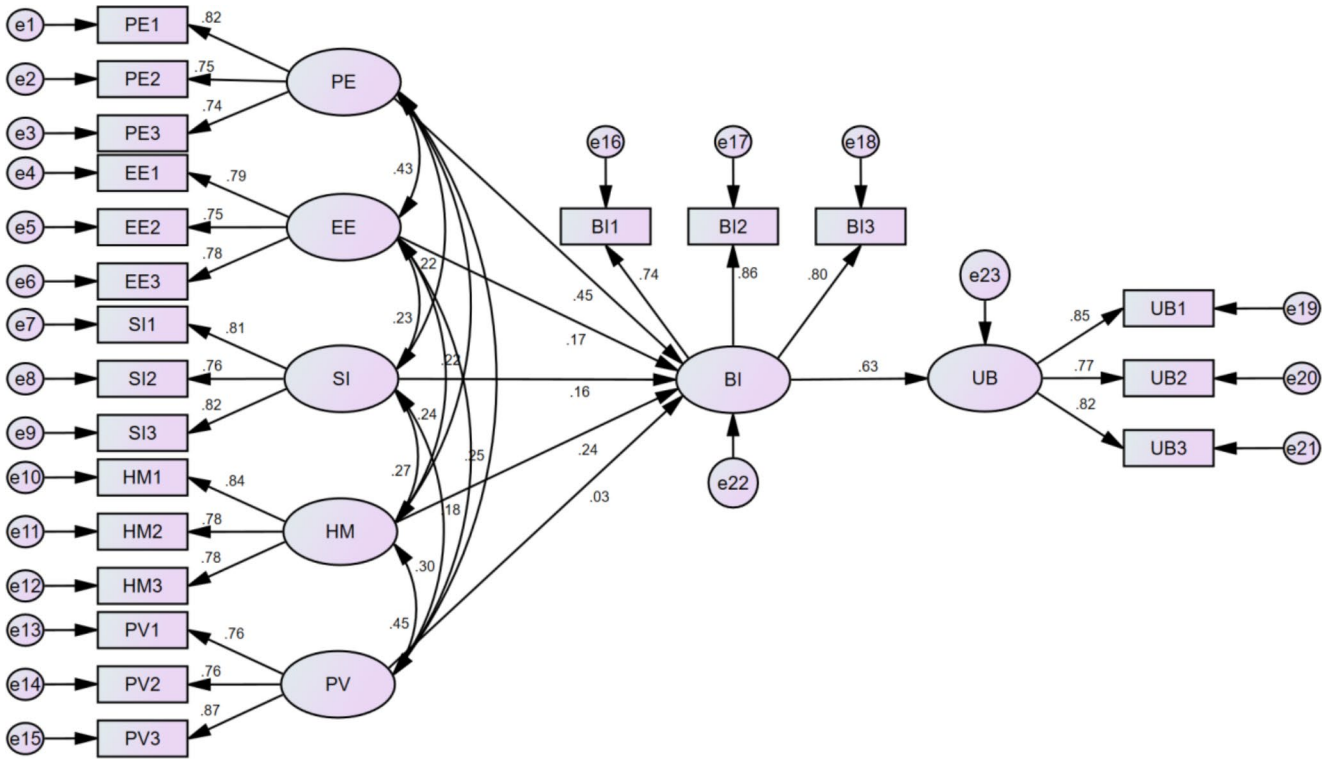

Fig. 2 Structural Equation Model

Table 4 Moderation effects analysis of habit and personal innovativeness

Variable	Model 1		Model 2		Model 3		Model 4	
	β	t	β	t	β	t	β	t
gender	−0.039	−0.874	−0.014	−0.365	−0.025	−0.683	−0.025	−0.712
age	0.049	1.047	0.029	0.721	0.034	0.899	0.049	1.368
edu	0.034	0.732	0.030	0.755	0.037	0.971	0.031	0.860
BI			0.508	13.278***	0.358	8.718***	0.311	7.757***
H					0.216	5.470***	0.184	4.836***
PI					0.225	5.972***	0.236	6.538***
BI*H							−0.169	−5.217***
BI*PI							0.187	5.139***
R^2	0.006		0.263		0.344		0.409	
$\triangle R^2$	0.006		0.257***		0.081***		0.065***	
F	0.947		45.035***		44.002***		43.339***	

*$p<0.05$, **$p<0.01$, ***$p<0.001$

impacts UB (β=−0.169, $p<0.001$), indicating that H plays a negative moderating role in the relationship between BI and UB. On the other hand, the interaction term between BI and PI has a significantly positive impact on UB ($\beta=0.187$, $p<0.001$), signifying that PI plays a positive moderating role in the relationship between BI and UB.

To further illustrate the moderating effects of H and PI on the relationship between BI and UB, this study conducted a simple slope analysis and presented moderation effect plots to depict the direction and strength of the moderating effects visually. Figures 3 and 4 show that when H has a low value, the positive impact of BI on UB is relatively significant ($\beta=0.480$, $p<0.001$). Conversely, when H has a high value, the positive impact of BI on UB is relatively tiny ($\beta=0.142$, $p<0.05$), confirming the hypothesis. Similarly, when PI has a low value, the positive impact of BI on UB is relatively tiny ($\beta=0.124$, $p<0.05$). However, when PI has a high value, the positive impact of BI on UB is relatively significant ($\beta=0.498$, $p<0.05$), once again confirming the hypothesis.

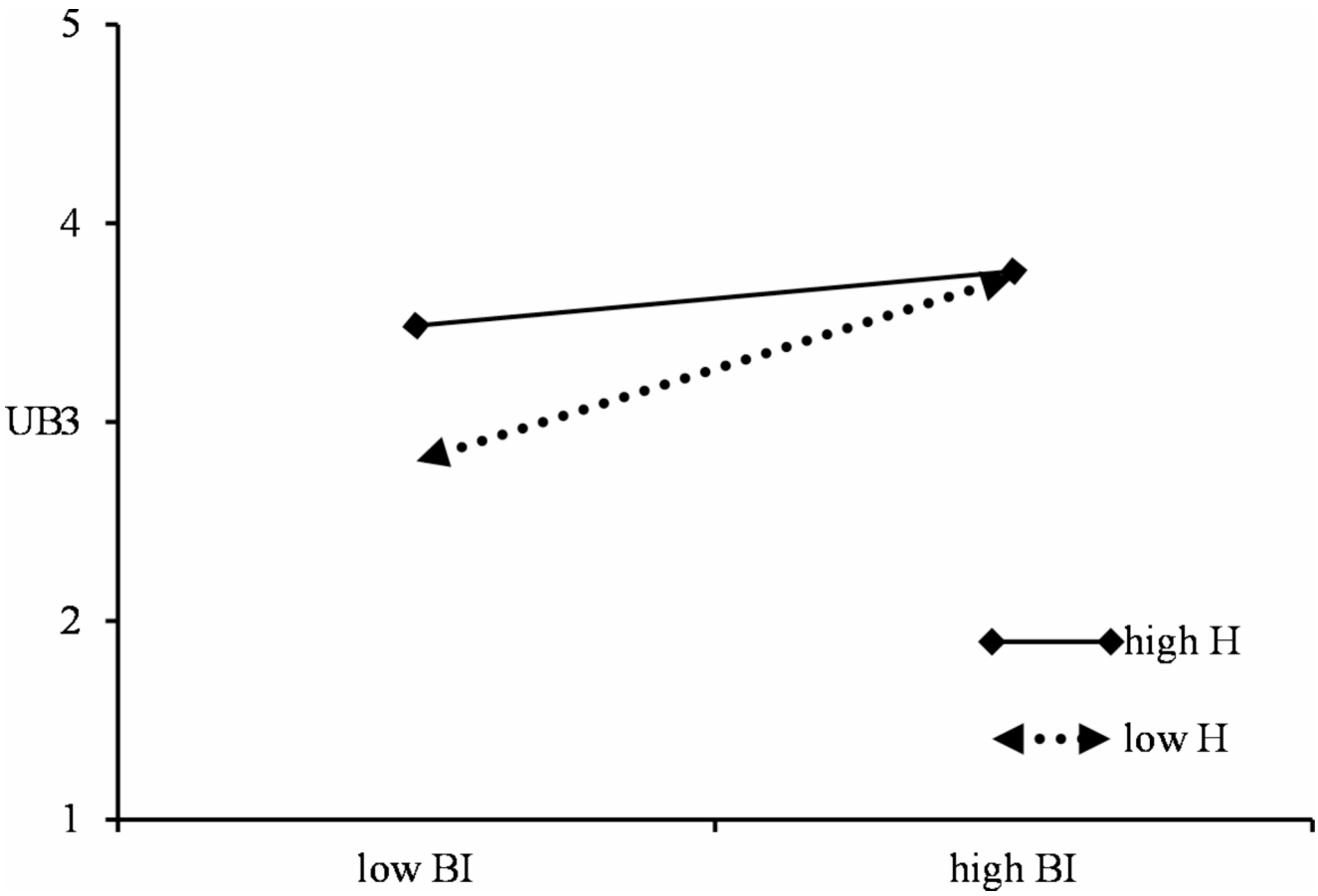

Fig. 3 Moderating Effect of Habit on the Relationship between Behavioral Intention and Usage Behavior

Results

Through an in-depth analysis of the data presented above, this study's proposed model demonstrates good reliability, validity, and significant correlations among the variables. Notably, except for hypothesis 5 not receiving support, the remaining hypotheses were validated in hypothesis testing. Specifically, the impact of Price Value on the intention of university students to use ChatGPT did not show a significant effect. This result diverges from the mainstream views in current academic research. Perhaps this can be explained by the variable of Price Value in this study, which selected ChatGPT 4.0 as the subject of inquiry. At the same time, most university students prefer the free ChatGPT 3.5 version. For these users, the free version is sufficient to meet their learning and daily needs, and thus, the perceived Price Value for ChatGPT 4.0 may need to be clarified. This finding aligns with Vafaei-Zadeh et al.'s (2018) study.

The Habit was found to have a negative moderating effect in this study, impacting the relationship between Behavioral Intentions and Usage Behavior. This could be attributed to university students forming Habit of using other GenAI products outside ChatGPT. Due to the behavioral similarity between ChatGPT and other GenAI products, when university students develop an intention to use, the negative moderating effect of Habit weakens their continuous Usage Behavior. This conclusion aligns with the results of Limayem and Cheung (2011) study, providing further understanding of the role of Habit in the mechanisms behind GenAI usage.

In contrast to Habit, Personal Innovativeness exhibits a positive moderating effect between Behavioral Intentions and Usage Behavior. This may be associated with university students' generally adventurous and trial-oriented personal characteristics. When individuals possess a higher level of Personal Innovativeness, they are more likely to accept and actively engage in GenAI usage, thereby promoting sustained Usage Behavior. This finding aligns with the results of Ratten (2014) study, providing crucial insights into the positive impact of Personal Innovativeness in GenAI usage.

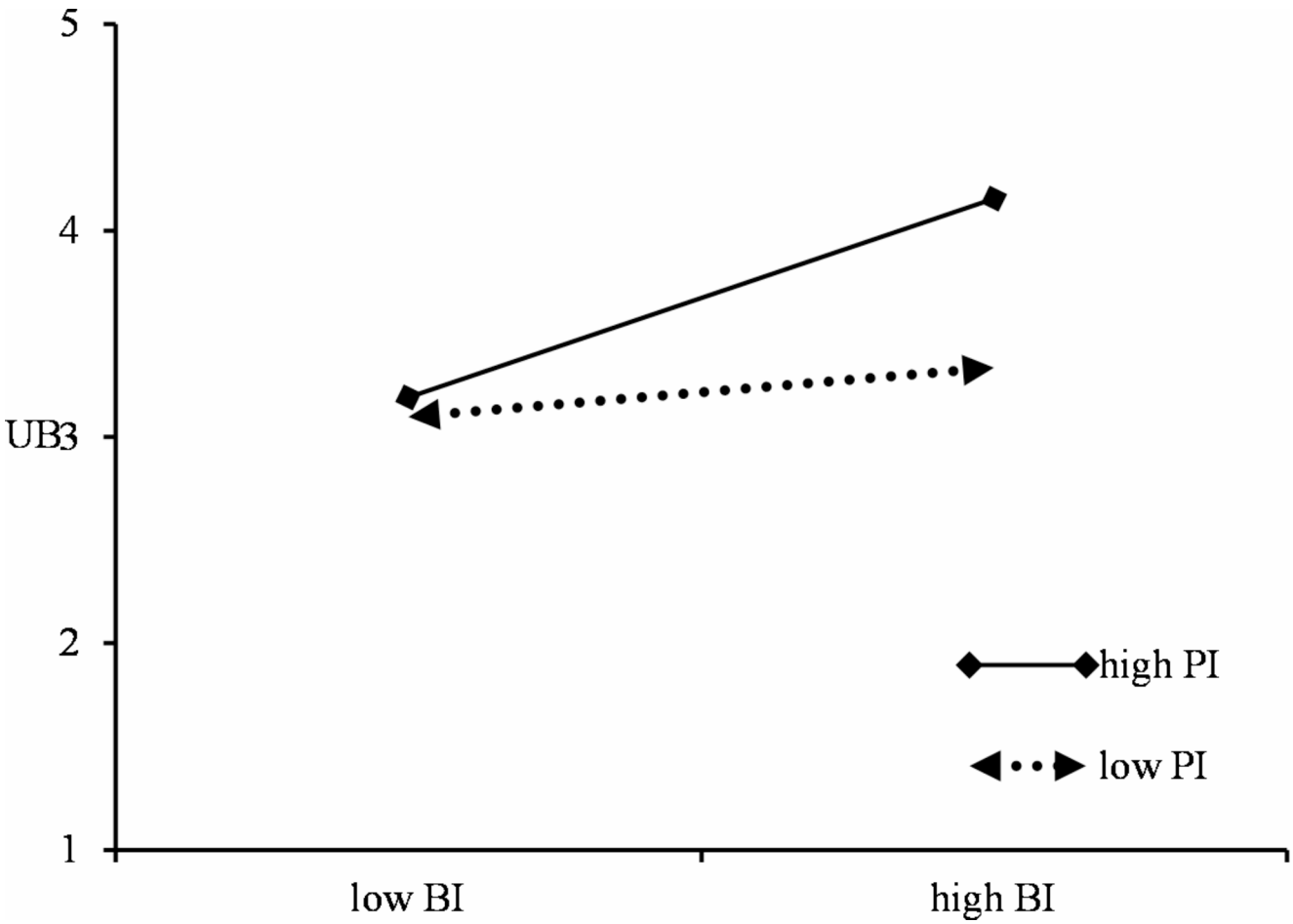

Fig. 4 Moderating Effect of Personal Innovativeness on the Relationship between Behavioral Intention and Usage Behavior

Discussion

This study enhances the understanding of university students' GenAI Usage Behavior by incorporating Habit and Personal Innovativeness into the investigation of GenAI Behavioral Intentions. Compared to previous research, this study provides a more comprehensive and in-depth explanation of university students' GenAI Usage Behavior. The unique aspect of this study lies in examining how Habit, as stable behavioral patterns, and Personal Innovativeness, as a proactive exploration capability, jointly moderate the intention to use GenAI. Furthermore, by taking ChatGPT as a case study, this research exhibits strong operability and practicality in empirical studies. ChatGPT, as a representative GenAI, enjoys widespread applications across various domains, making the research findings more generalizable and beneficial for designing and improving other GenAI products. Most importantly, this study offers valuable insights into the factors influencing GenAI Behavioral Intentions, providing targeted guidance for GenAI product designers and decision-makers. By understanding users' behavioral patterns and proactive exploration capabilities, these stakeholders can enhance user experience and trust, thereby promoting the broader application of GenAI technology.

The significance and contribution of this study lie in the in-depth analysis of GenAI usage intention, providing a crucial reference for advancing the sustainable development of GenAI technology. Firstly, comprehending the roles of Habit and Personal Innovativeness in GenAI usage contributes to optimizing GenAI design and enhancing user engagement and satisfaction. Secondly, through constructing a model of factors influencing usage intention, this study not only explains GenAI user behavior but also offers specific recommendations for the future improvement of GenAI products. In practical terms, this research aids in addressing issues encountered in the use of GenAI products like conversation capacity limitations, ethical considerations, intellectual property rights, data biases, etc., thereby increasing user trust in GenAI and promoting more practical applications in education, research, consulting, entertainment, and other domains. Through a profound exploration of university students' GenAI usage intention, this paper provides valuable

theoretical support for advancing artificial intelligence technology's sustainable development and societal application.

Limitations and prospects

While this study has achieved valuable findings in revealing the influencing factors of GenAI usage intention, some limitations should be acknowledged and further investigated. Firstly, the empirical research primarily focused on ChatGPT as a representative in the GenAI field. Given the rapid advancements in GenAI products and technologies, there may be emerging models with different characteristics. Therefore, the conclusions of this study may have specific limitations when applied to other GenAI products. Future research could expand the sample scope to encompass a broader range of GenAI products, validate the model's generalizability, and gain a more comprehensive understanding.

Secondly, although this study attempted to delve into the mechanisms behind GenAI usage intention by introducing Habit and Personal Innovativeness as moderating variables, considering other potential moderating variables still appear limited, future research could further expand the range of moderating variables to provide a more comprehensive understanding of the formation process of GenAI usage intention. Additionally, this study employed a survey questionnaire, which, to some extent, relies on subjective responses from the respondents. Future research might consider incorporating actual usage data or experimental designs to enhance the objectivity and reliability of research outcomes.

Despite these limitations, the innovation and depth of this study still provide valuable insights for research in the GenAI field. For GenAI product designers and decision-makers, this research offers more targeted guidance, helping them better understand the motivations behind user behavior, thereby optimizing product design and advancing the further application of GenAI technology. Looking ahead, with the continuous development of GenAI technology, its application scenarios and potential impacts will continue to expand. Future research can focus on the application of GenAI in various fields and its impact on individuals, organizations, and society. At the same time, emphasis should be placed on interdisciplinary cooperation and exchange to comprehensively reveal the usage behavior and impact mechanisms of GenAI through the integrated application of theories and methods from different disciplines.

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Data availability The datasets analyzed in the current study are not publicly available due to restriction of research foundation but are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.No funding was received to assist with the preparation of this manuscript.

Informed consent Exploring the Usage Behavior of Generative Artificial Intelligence: A Case Study of ChatGPT with Insights into the Moderating Effects of Habit and Personal Innovativeness.

Ethics disclaimers 1.Each author has made substantial contributions to the conception, design, analysis, and interpretation of data for this study.2.The manuscript has not been published previously and is not currently being considered for publication elsewhere3.All experiments comply with the ethical standards of our institution and with national and international guidelines.4.All data presented in this manuscript are truthful and accurate, and all sources have been appropriately acknowledged.5.The authors have full control of all primary data, and agree to allow: Current Psychology to review the data if requested.6.The authors have read and approved the final version of the manuscript.7.There are no financial or non-financial competing interests to declare.
The study is compliant with Qianli Wu institution's and Korean ethical standards. It conforms to the 1964 Declaration of Helsinki and its later addenda, and informed consent was obtained from all participants.

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